

A Prudent Question Is One-Half of Wisdom

WHY.

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Abstract

Language models are trained to answer. A system that instead *asks* — that puts a small set of next questions in front of a person and lets them choose — generates evidence that exists in no text corpus: revealed preference over alternatives that were legible, contemporaneous, and declined. We argue that this evidence, not the model producing the questions, is the durable asset in such a system, and we give the mechanism that accumulates it.

Slate selection is an experimental design problem. We show that the obvious objective, maximising the entropy of the induced choice, is wrong in a specific and correctable way: it cannot distinguish a slate the system knows too little about from one whose outcome is inherently arbitrary, and so rewards manufactured confusion. The correct objective is the expected reduction in uncertainty about the preference parameters — mutual information between parameters and outcome. In simulation the two criteria select different slates in 92.9% of instances, and a slate of maximal choice entropy can carry exactly zero information.

The observations accumulate into a **Curiosity Graph**: a directed weighted graph whose vertices are canonical questions and whose edges record that one was declined in favour of another. We rank it with a PageRank-style recursion in which a question gains standing by defeating questions that themselves have standing, and feed that standing back into the utilities selecting the next slate. Each turn contributes $k-1$ ordered pairs rather than one click, and the declined options are not supplementary evidence: they are the term that makes the choice likelihood computable at all.

1 The Engine That Asks

A model that answers a question is judged on the answer. A system that hands a person three questions and asks them to pick one is doing something else, and the standard by which it should be judged is not obvious.

The default standard is engagement: offer the continuations most likely to be taken. We argue this is not merely imperfect but self-defeating for a system meant to learn what a person wants. A slate of alternatives is an experiment; its outcome is a measurement; and what it measures is the uncertainty it resolves. A policy that always presents the option it is most confident about learns, when that option is taken, that it was right about something it already knew. The regime of greatest confidence becomes the regime of least learning, and confidence is never corrected.

The remedy is to treat slate construction as experimental design — but the naive version of that remedy fails too, and understanding why is most of the technical content of this paper. Maximising the entropy of the predicted choice does not maximise learning, because choice entropy conflates two things: uncertainty the system could remove by observing (it does not yet know the person) and uncertainty it cannot (the person is close to indifferent). Three equally appealing questions produce maximal choice entropy and may teach nothing at all. Section 5 separates the two and gives the objective that isolates the first.

A second point concerns what is recorded. A conventional log stores what was chosen but not what it was chosen *over*, because in a feed or a page of results the alternatives are unbounded, unread, or both. A three-door board bounds them and makes them legible, so each turn yields $k-1$ ordered pairs — and in a conditional-logit model the declined options appear in the normalising term, without which the likelihood of the observed choice cannot be evaluated at all.

Accumulated across many people and turns, those comparisons form a graph over questions. We call it the Curiosity Graph and argue in Section 6 that it, rather than any model, is what a question engine is for: a record of what people found worth pursuing when the alternatives were spelled out. Pretraining on text does not produce it, because the observation is not in the text.

2 The Problem Is Old, The Data Is New

Choosing one alternative from a presented set to recover a latent utility is the subject of discrete choice theory; the conditional logit we use is McFadden's [1]. Estimating latent strengths from incomplete comparisons originates with Bradley and Terry [2], whose one-from-many

generalisation is the Plackett–Luce family. Ranking by the dominant eigenvector of an outcome matrix — the structure behind PageRank [3] applied to contests rather than links — is Keener's method [4]; the personalised teleport of Section 6.2 is topic-sensitive PageRank [11].

Choosing an experiment to maximise expected information about a parameter is due to Lindley [5]. The specific decomposition we use in Section 5.3, which separates uncertainty about the parameters from uncertainty inherent in the outcome, is Bayesian Active Learning by Disagreement [12]; for multinomial-logit designs the Fisher-information and Bayesian D-optimal alternatives are also established [13]. The online form, in which one must learn from comparisons while still serving the participant, is the dueling bandit problem of Yue and Joachims [6]. Recovering unbiased rankers from position-biased logged feedback is treated by Joachims et al. [7], whose propensity-weighted estimators we adopt rather than re-derive.

The nearest neighbour in current practice is preference modelling for language models, in which a reward model is a conditional-logit fit over annotator comparisons. It differs in the respect that matters here: its alternatives are constructed for a paid rater, whereas ours are read and declined by the person the ranking will serve, during a task they chose to undertake. Our companion report [8] argues the negative case for similarity-based retrieval and is assumed here.

3 An Instrument for Measuring Rejection

For each question a person asks, the system returns one compact answer and exactly $k = 3$ further questions. The person takes one, or leaves. The selection, its depth in the session and any later return are recorded.

What matters is not that three questions are shown but that the two not taken were *read and declined*: contemporaneous, legible, specific. The person knew what they were giving up.

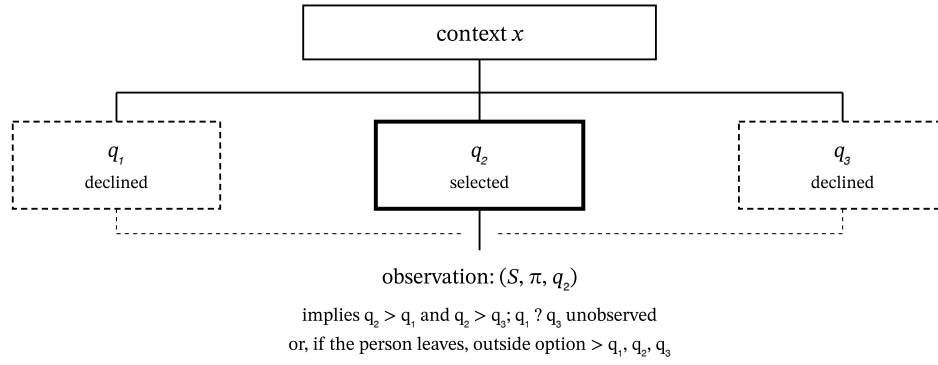


Figure 1: One turn. A context yields $k = 3$ generated questions in presentation order π . The selection implies $k-1$ ordered pairs, where $>$ denotes revealed preference. These are two implications of one observation, not two independent observations (Section 4.3). The relation between the declined pair is unobserved by construction. Leaving without selecting is a fourth outcome, not a missing row.

4 The Half of the Choice Everyone Discards

It is tempting to treat the declined options as context for the selection. The relationship runs the other way.

4.1 Rejections Make the Likelihood Computable

Write the conditional-logit log-likelihood of one observation, with S the slate and q the option taken:

$$\log P(q \mid S, x) = \theta(q \mid x) - \log \sum_{q' \in S} \exp \theta(q' \mid x) \quad (1)$$

The first term uses the chosen option. The second — the log-partition — ranges over the entire slate and is the only term carrying the alternatives. A log recording selections without their choice sets cannot evaluate it, and therefore cannot evaluate the likelihood at all. This is not a matter of degree: preference is a relation over alternatives, and a record of one side of a relation does not identify the relation however large it grows.

4.2 Every Update Runs Through the Declined

The same asymmetry governs learning. With $\theta(q \mid x) = w^\top \varphi(q, x)$, differentiating (1) gives the observed-minus-expected form

$$\nabla_w \log P = \varphi(q, x) - \sum_{q' \in S} P(q' \mid S, x) \varphi(q', x) \quad (2)$$

The expectation runs over the whole slate, so the declined options enter the update directly, weighted by their predicted probability. When those probabilities are near uniform the declined pair accounts for $(k-1)/k$ of the probability mass in that term. We state this as a statement about the mass and not about "the learning signal": the actual contribution to the update depends on the feature vectors φ , and two declined options whose features nearly coincide with the chosen one contribute little however large their probability.

4.3 Two Implications, One Observation

A three-way choice implies $q > q'$ for each declined q' , but these are two consequences of a single act, not two independent draws. They share the same context, the same slate, and the same person at the same moment, and their errors are correlated by construction. Counting 265,000 resolved slates as 530,000 comparisons is correct bookkeeping for the size of the edge set; it is not a claim to twice the independent information, and standard errors must be clustered at the level of the observation rather than the pair. Section 8 reports both counts for this reason.

4.4 Leaving Is an Answer

Leaving without selecting is a fourth outcome with utility θ_0 , entering the denominator of (1) alongside the k candidates. Recording it as an absent row rather than an outcome makes the model treat the least bad of three poor options as a preference, and discards the strongest available evidence about a bad slate. Section 8.2 bounds how often it occurs on the deployed system, and explains why the present instrumentation can only bound it.

5 Choosing the Slate as an Experiment

5.1 A Model of Choice

Let x denote the context, $C(x)$ a pool of candidate next questions, and $\theta(q | x)$ a real utility. A slate is $S \subseteq C$ with $|S| = k$. Selection follows

$$P(q | S, x, \theta) = \frac{\exp \theta(q | x)}{\sum_{q' \in S} \exp \theta(q' | x)} \quad (3)$$

We are Bayesian about θ : after data D the system holds a posterior $p(\theta | D)$, represented in practice by an ensemble, a Laplace approximation, or dropout samples.

5.2 The Objective That Rewards Confusion

The obvious criterion is the entropy of the predicted choice,

$$\mathcal{H}(Y | S, x, D) = - \sum_{q \in S} \bar{P}(q) \log_2 \bar{P}(q), \quad \bar{P}(q) = \mathbb{E}_{\theta|D} [P(q | S, x, \theta)] \quad (4)$$

Proposition 1. $\mathcal{H}(Y | S, x, D) \leq \log_2 k$, with equality if and only if $\bar{P}$ is uniform on S .

Proof. Appendix A. □

Maximising (4) is nonetheless a mistake, and the reason is that choice entropy sums two kinds of uncertainty which have opposite implications for learning. *Epistemic* uncertainty is the system's ignorance of θ and is removed by observing. *Aleatoric* uncertainty is variability that remains even when θ is known exactly — the person is genuinely close to indifferent — and no observation removes it.

A slate of three options the system knows, with confidence, to be equally appealing attains the maximum of (4) and teaches nothing whatever. Worse, such slates are cheap to manufacture: three near-paraphrases are indistinguishable by construction. An objective that rewards them is an objective that rewards confusion.

Table 1: Two slates under a posterior represented by samples of θ . Slate A: the system is confident the three utilities are equal. Slate B: half the posterior mass favours q_1 , half favours q_3 . Choice entropy prefers A; the mutual information of (5) prefers B, and assigns A exactly zero.

Slate	$\mathcal{H}(Y)$	$\mathbb{E}_{\theta}[\mathcal{H}(Y \theta)]$	$\mathbb{I}(\theta; Y)$
A — known to be equal (aleatoric)	1.585	1.585	0.000
B — posterior disagrees (epistemic)	1.404	0.636	0.768

5.3 Measure Disagreement, Not Ambiguity

The quantity we actually want is the expected reduction in uncertainty about θ from observing the choice — the mutual information between parameters and outcome, which decomposes as total predictive entropy minus expected conditional entropy [5, 12]:

$$\mathbb{I}(\theta; Y | S, x, D) = \mathcal{H}(Y | S, x, D) - \mathbb{E}_{\theta|D} [\mathcal{H}(Y | S, x, \theta)] \quad (5)$$

The subtracted term is the aleatoric part: the entropy that would remain if θ were known. What survives is disagreement *among the models the posterior still admits*. The criterion is therefore not "offer three questions you cannot choose between" but **offer three questions your surviving hypotheses would rank differently**.

With M posterior samples $\theta^{(1)}, \dots, \theta^{(M)}$, both terms are direct averages over the samples and (5) costs $O(Mk)$ per slate. In simulation over 4,000 instances with $n = 8$ candidates and $M = 12$

samples, the slate maximising (4) differed from the slate maximising (5) in **92.9%** of cases. The distinction is not a refinement; it changes the decision almost always.

5.4 The Rule in One Line

Maximising (5) alone would return three irrelevant questions about which the posterior happens to disagree. The objective must therefore be constrained rather than blended: a weighted sum of relevance, information, novelty and diversity is dominated by whichever term carries the largest scale, and its weights cannot be tuned against one another without a criterion this setting does not supply.

$$\begin{aligned}
 S^* = \operatorname{argmax}_{S \subseteq C(x), |S|=k} \quad & \mathbb{I}(\theta; Y \mid S, x, D) \quad \text{subject to} \\
 \text{relevance floor} \quad & \mathbb{E}_\theta[\theta(q \mid x)] \geq \tau \quad \text{for all } q \in S \\
 \text{divergence} \quad & d(q, x) \geq \delta \quad \text{for at least one } q \in S \\
 \text{non-redundancy} \quad & \operatorname{sim}(q_i, q_j) \leq \rho \quad \text{for all } i \neq j
 \end{aligned} \tag{6}$$

Equation (6) is the whole of the selection policy. The floor τ keeps every option worth reading; δ reserves a slot for a direction departing from the current line of reasoning; ρ blocks the near-paraphrase degeneracy of Section 5.2. Note that ρ is doing less work here than it would under choice entropy: three paraphrases are aleatorically uncertain, so (5) already scores them near zero. The constraint remains as a guard rather than as the primary defence, which is one practical argument for the corrected objective.

5.5 Why the Clever Shortcut Fails

An earlier version of this work proposed an efficiency result: with candidates sorted by utility, the entropy-maximising slate is contiguous in that order, so the search reduces to a linear scan of windows. The claim is true for unconstrained choice entropy (Appendix A, Proposition 2) and *it does not support the selector of (6)*.

The constraints break it. Take five candidates with utilities 5, 4, 3, 2, 1 and suppose the redundancy ceiling invalidates every adjacent pair. The feasible slates are then those with no two adjacent indices — of which {1, 3, 5} is the only one — while every contiguous window contains an adjacent pair. A window scan rejects all three candidate windows and returns nothing, although a feasible slate exists. The result therefore cannot be used to justify the algorithm, and we withdraw it as an operational claim.

This costs nothing in practice. The candidate pool is small by construction, and exhaustive enumeration of feasible triples is trivial: $C(12, 3) = 220$ and $C(40, 3) = 9,880$, each evaluated in

$O(Mk)$. We therefore score every feasible slate and keep the best.

Algorithm 1 $\text{SelectSlate}(C, x, k, \tau, \delta, \rho, \{\theta^{(m)}\})$

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1   $F \leftarrow \{ q \in C : \text{mean\_m } \theta^{(m)}(q|x) \geq \tau \}$            relevance floor
2   $\text{best} \leftarrow \perp$  ;  $I^* \leftarrow -\infty$ 
3  for each  $S \subseteq F$  with  $|S| = k$ :            $C(|F|, k)$  triples
4      if not  $\text{divergent}(S, x, \delta)$ : continue
5      if not  $\text{distinct}(S, \rho)$ : continue
6      for  $m$  in  $1..M$ :  $p^{(m)} \leftarrow \text{softmax}(\theta^{(m)}[S])$ 
7       $\bar{p} \leftarrow \text{mean\_m } p^{(m)}$ 
8       $I \leftarrow H(\bar{p}) - \text{mean\_m } H(p^{(m)})$            eq. (5)
9      if  $I > I^*$ :  $I^* \leftarrow I$  ;  $\text{best} \leftarrow S$ 
10 if  $\text{best} = \perp$ : relax  $\rho$  by one step and retry       feasibility fallback
11 return  $\text{shuffle}(\text{best})$                              randomised order

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Line 10 matters: a constrained selector can have an empty feasible set, and a policy that can return nothing is not deployable. Line 11 is not cosmetic — presentation order must be randomised per impression or position effects and preference are not separately identified (Section 8.4).

5.6 Against Ranking by Likelihood

We compare (6) against a *likelihood-ranked* baseline policy: select the k candidates of highest expected utility, maximising $\sum_{q \in S} \mathbb{E}[\theta]$ rather than (5). This is the natural policy for a system optimising for selection, and it is a reasonable stand-in for asking a language model for its most likely follow-up questions — though we are careful not to assert that any particular model's token likelihood coincides with predicted human utility. That correspondence is an empirical question we have not tested, and H2 of Section 9 tests the policies rather than the claim.

Under the simulation of Section 5.3 the top- k slate maximised choice entropy in 11.3% of instances. Both this figure and the 92.9% above are properties of the simulated distribution — Gaussian utilities, eight candidates, twelve posterior samples — and are reported to show that the criteria diverge materially, not as universal constants. What is not distribution-specific is the direction: the two agree only where the posterior is already indifferent among the top candidates, and diverge as the system grows confident.

6 The Curiosity Graph

Everything so far concerns a single turn. The asset is what the turns accumulate.

6.1 What Accumulates

Let V be a set of canonical questions — generated text mapped into shared identities, so that two turns asking the same thing in different words reach the same vertex. Each resolved slate contributes $k-1$ directed edges, from each declined question to the one taken. Let $w(j, i)$ be the accumulated weight of edge $j \rightarrow i$, meaning j was declined in favour of i . The pair (V, w) is the Curiosity Graph.

Three properties distinguish it from a click log, a knowledge graph or a personal memory. It is *comparative*: an edge is the outcome of a contest between named alternatives, not a co-occurrence. It is *population-level*: every person's comparisons inform every other person's ranking, so the structure compounds across users rather than per user. And it is *counterfactual*: the edge exists only because something was passed over.

6.2 Standing Earned by Defeating the Strong

Utilities in (3) are local and context-conditional. We also want a global standing for a question, independent of the context in which it last appeared, and the natural criterion is recursive: a question has standing if it defeats questions that themselves have standing. This is the structure of PageRank [3] over a tournament rather than a link graph, and it is Keener's method [4]:

$$CR(i) = (1 - d) B(i) + d \sum_{j \rightarrow i} \frac{CR(j) \cdot w(j, i)}{\sum_k w(j, k)} \quad (7)$$

with $d \in (0,1)$ a damping constant and B a teleport distribution over V summing to one. Uniform B gives a global ranking; setting B to a person's history gives the topic-sensitive variant [11]. Two mechanical points carry over from the link setting. A question never declined has no outgoing edges, so its mass must be redistributed explicitly, exactly as PageRank handles dangling nodes. And (7) is well defined only on a connected graph: if canonicalisation fails and each slate forms its own component, the sum contributes nothing and CR collapses to B .

Estimation and presentation are distinct jobs and we keep them apart. The choice model of (3) estimates, because Section 5 needs calibrated probabilities; the recursion (7) presents, because a ranking needs one number per question. CR re-enters estimation as a feature through (10), closing the loop.

6.3 The Asset a Better Model Cannot Buy

A competitor with a better language model still has no comparisons, and the reason is structural rather than incidental.

A pretrained model holds a prior over text continuations. From it one can guess what question a person might ask, because such questions appear in text. What does not appear in text is

which of three specific, simultaneously presented alternatives a particular person took at a particular depth of a particular inquiry, having read and declined the other two. That observation is generated only by an instrument that presents bounded, legible alternatives and records the outcome. The slates themselves are visible to any visitor and could be copied; what cannot be copied is which alternative each person took, because that is never rendered. The asset is the choices rather than the questions, and it is unavailable to a competitor for the same reason it is absent from any text corpus: the counterfactual was never written down.

- *It compounds.* Each resolved slate adds edges, refines the posterior through (2) and updates CR through (7); the updated CR improves the next slate through (10). The asset improves the mechanism that produces it.
- *It yields more per turn than a click log.* The same n turns yield n observations that are each evaluable under (1), which a selection log is not — subject to the dependence noted in Section 4.3.
- *Its marginal cost is a by-product.* The comparisons are produced by people pursuing their own questions, so acquisition cost is the inference cost of serving them — $\$2.5 \times 10^{-4}$ per observation in Section 8. We do not present this as a like-for-like comparison with commissioned annotation, because it is not one: a rater's judgement carries screening, instruction and quality control that a visitor's click does not. What the instrument buys is volume and ecological validity at negligible marginal cost; what it gives up is any guarantee that a particular observation was attentive.

At the deployed unit economics, a system serving 10^9 questions accumulates on the order of 10^9 resolved slates and 2×10^9 ordered pairs for an inference cost near $\$2.5 \times 10^5$. It is that graph, and not the generator in front of it, that a question engine is for.

7 The System as One Loop

The components assemble into a single loop: a generator proposes candidates; a scorer assigns utilities under the current posterior; the selector applies (6); the presenter randomises order; the recorder writes the outcome; the estimator refits; the ranker recomputes CR; and CR returns to the scorer.

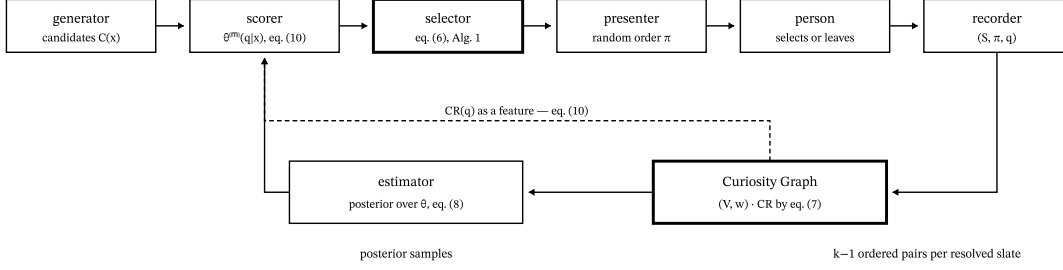


Figure 2: The loop. Solid arrows carry a single turn; the dashed arrow carries accumulated structure back into scoring. The two heavy boxes are the contributions of this paper — the selection rule of Section 5 and the graph of Section 6.

The posterior over θ is maintained by penalised maximum likelihood with a Laplace approximation, or equivalently by an ensemble:

$$\mathcal{L}(w) = \sum_t \log P(q_t | S_t, x_t, w) - \lambda \|w\|^2 \quad (8)$$

whose gradient is (2) summed over turns, with standard errors clustered by observation per Section 4.3. Position enters the feature map as a covariate,

$$\varphi(q, x) = [\text{content} ; \text{role} ; \text{promise} ; \text{depth} ; \text{domain} ; \text{position}] \quad (9)$$

and the global standing of a question enters the utility additively in log space, so a question with a strong record across the population starts higher in an unfamiliar context:

$$\theta(q | x) = w^\top \varphi(q, x) + \gamma \log \text{CR}(q) \quad (10)$$

Equations (3)–(10) are the entire system: a choice model, an information objective, three constraints, a likelihood, a graph recursion, and one coupling constant γ setting how far population evidence overrides local context.

One property of the assembled loop belongs in the text rather than in a footnote. Equation (10) makes θ depend on CR, while CR is computed from edges generated by slates that were themselves selected using θ . The loop is closed, and its feedback runs in the obvious direction: a question with high standing scores higher, is therefore more likely to be selected into a slate, and on being taken accrues further standing. This is the rich-get-richer dynamic familiar from link graphs, with the selection policy inside the loop rather than outside it.

We have not shown that it converges. The damping constant d bounds the influence of the recursion at each step and the teleport term B keeps mass on questions with no incoming edges, but neither is a stability result for the composed system. Nor is the objective of Section 5 obviously protective here: its anti-narrowing property was argued for the entropy of a single

slate, not for the trajectory of CR under repeated selection, and a system concentrating standing on a shrinking set of questions could satisfy (6) at every individual turn while narrowing over months.

The diagnostic is cheap even where the proof is absent: track the entropy of the CR distribution across the vertex set and treat a monotone decline in it as a fault condition rather than as convergence. Bounds on γ , and any stability result for the coupled system, we leave open.

8 What the Deployment Shows

8.1 Six Hundred Thousand People, Sixty-Seven Dollars

Over a 72-hour period of organic traffic in August 2026 the reference system served 677,000 unique visitors and processed 317.8M tokens for approximately \$67 of inference.

Table 2: Measured quantities from the 72-hour window (upper block) and quantities derived under $k = 3$ (lower block). Derived rows assume each completed request resolves one slate. Note the distinction between independent observations and the ordered pairs they imply (Section 4.3).

Quantity	Value	Basis
Unique visitors	677,000	analytics, 72 h window
Tokens processed	317.8 M	provider accounting
Inference cost	\$67	provider accounting
Completed requests	$\approx 265,000$	at $\approx 1,200$ tokens/request
Requests per visitor	0.39	derived
Independent observations	$\approx 265,000$	one per resolved slate
Ordered pairs implied	$\approx 530,000$	$k-1$ each, dependent
Cost per visitor	$\$9.9 \times 10^{-5}$	derived
Cost per observation	$\$2.5 \times 10^{-4}$	derived
Cost per ordered pair	$\$1.3 \times 10^{-4}$	derived

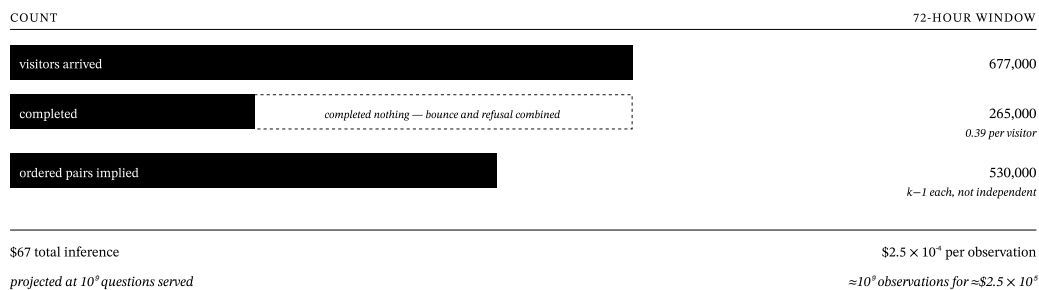


Figure 3: The 72-hour window as a yield funnel. Solid bars are proportional to the counts in Table 2; the dashed remainder on the second row is arrivals that completed no question, which is an upper bound on the outside option of Section 4.4 rather than a measurement of it, since it also contains visitors who left before a slate was rendered (Section 8.2). The third bar exceeds the second because each slate implies $k-1$ pairs, but those pairs are dependent and the second bar is the count that governs statistical power.

8.2 Most Arrivals Complete Nothing

At 0.39 requests per visitor, most arrivals completed no question. We report this because it bounds every claim here: the observations rest on a minority of the traffic.

It does not, however, measure the outside option of Section 4.4, and an earlier draft of this report was careless on the point. The 0.61 of arrivals completing nothing contains at least two populations the present instrumentation cannot separate: those who left before any slate was rendered, and those who read a slate and declined all of it. Only the second is the outside option. Bounce and refusal are different behaviours with different implications, and treating them as one would inflate the very quantity Section 4.4 argues is being discarded.

What the figure establishes is therefore an upper bound — the outside option occurs on at most 0.61 of arrivals, and plausibly on far fewer. The decomposition requires exactly the instrumentation Section 8.4 lists as missing: a slate-rendered event paired with a slate-resolved event, after which the rate is measured rather than bounded. That the system cannot currently tell a refusal from a bounce is itself the argument for recording the outcome.

8.3 A Shared Vocabulary Already Exists

Each generated question carries typed fields from small closed vocabularies: a role from six values, a promise from three, and a domain. Roles map onto promises two-to-one.

Table 3: The typed vocabulary emitted with every question. These identities are shared across turns by construction, so a graph over them is connected without any canonicalisation of question text.

promise	roles	reading
microscope	deepen, origin	go beneath the answer
trapdoor	contradiction, surprise	go against the answer
telescope	consequence, pattern	go beyond the answer

This matters for (7). Question text is generated fresh each turn and does not repeat, so a graph over raw strings is disconnected — one component per slate — and CR collapses to the teleport vector. A graph over the typed vocabulary is connected immediately and supports estimation of role and promise utilities from telemetry already collected. It is coarse: it records that a question was an origin question, not what it was about. It is the part of the Curiosity Graph estimable today, and canonicalisation of question text is the work that widens it.

8.4 What Is Not Yet Built

Four gaps determine the shape of the evaluation. Slate order is the generator’s output order and is not randomised, so position effects and preference are not separately identified; historical estimates require the propensity weighting of [7] and remain descriptive. No posterior over θ is maintained, so the system does not implement (6). Question text is not canonicalised, so the graph available today is the typed one of Section 8.3. And the outside option is not recorded as an outcome.

The consequence is convenient for evaluation and unflattering for the product: **the system as deployed implements the likelihood-ranked baseline of Section 5.6**. It is the control condition against which rule (6) is to be tested.

9 What Would Prove This Wrong

We state four hypotheses in increasing order of strength with the observation that would refute each. Splits are at the level of the person, never the event; standard errors are clustered by observation; minimum effect sizes are fixed before running; every reported difference carries an interval. Engagement measures — session length, depth reached, slates opened — are recorded as diagnostics and excluded from every success criterion, so a variant raising engagement while lowering task outcome is recorded as a failure.

H1 (information). Slates chosen by (6) produce a larger reduction in posterior uncertainty about θ per turn than likelihood-ranked slates at equal candidate cost. *Refuted if* posterior con-

traction does not separate, indicating the model is too poorly calibrated for (5) to bind.

H2 (learning rate). A model trained on slates chosen by (6) attains lower held-out predictive loss than one trained on likelihood-ranked slates at an equal number of observed turns. *Refuted if* the loss curves coincide. This is the claim that a question engine learns faster than a selection-maximising one, and it is the central hypothesis of this paper.

H3 (transfer). Utilities and CR fit on one population reduce held-out loss for people absent from the training data, and for subject areas absent from it. *Refuted if* they do not, in which case the method is personalisation requiring per-person history rather than a population asset, and the argument of Section 6.3 fails with it. The two axes are separate and the subject axis is the harder one: a graph that transfers between people but not between topics is an asset per domain rather than a general one. This is the moat test.

H4 (outcome). Conditioning an agent's next action on the fitted utilities improves task outcomes — completion, correction rate, whether the work was accepted — against the strongest baseline from H2 at matched candidate-set size. *Refuted if* outcomes do not improve while H1–H3 pass, establishing that the system predicts selection well and improves nothing. This is the outcome the programme most needs to exclude, because it is consistent with every measurement above it.

H5 (persistence). What a person declines predicts their distal choices better than what they selected. The mechanism we conjecture is that a selection is a draw from a context-dependent distribution — it varies with what happened to be on offer and what the person had just read — whereas the decline set is an intersection taken across many contexts, and intersections are more stable than draws. A person representation built from declines should therefore show higher test–retest stability and better predict selections weeks later. *Refuted if* selection-derived and decline-derived representations predict distal choice equally well. The control that decides this is severe: declines outnumber selections ($k-1$) to one, so the comparison must be made at a matched number of observations rather than a matched number of turns, or any advantage is an artefact of sample size. Of the five hypotheses this is the one whose confirmation would be genuinely surprising, and it is testable with the instrument exactly as described.

10 What Success Must Leave Behind

Section 9 states what would refute the thesis. This section states the converse: the artefacts that must exist for the programme to have been worth running. They are deliberately separable, because some survive the failure of others.

- **A corpus of bounded choice with recorded rejections.** Choice sets, presentation order, the selection, and the outside option, at population scale — the observation Section 4 argues is missing everywhere else. This is the most durable of the five: datasets outlive the

methods built on them, and this one is citable whether or not the objective of Section 5 wins. It also carries the heaviest obligations, since Section 11 notes that a decline vector is more disclosive than a selection vector.

- **A rejection-informed objective that beats strong baselines.** Not a baseline chosen after the fact: the comparison set must be fixed in advance and must include a well-tuned likelihood-ranked policy, since that is what the deployed system already does.
- **Evidence of transfer across people and across subjects.** Both axes, reported separately, per H3.
- **A canonicalisation method for generated alternatives.** Mapping freely generated questions into shared identities stable enough to support a graph, with the merge threshold fit rather than assumed. We expect this to be the most reusable technical contribution here, because the problem is not specific to question engines: anyone learning preferences over model-generated alternatives must decide when two generations are the same alternative, and the comparison graph is disconnected until they do.
- **A result that changes what people expect.** H5 is the candidate: that the questions a person turned down describe them better than the questions they took.

The first and fourth are the ones that survive if the objective of Section 5 turns out not to beat its baselines. A corpus and a canonicalisation method are useful to anyone working on preference learning over generated alternatives; the ranking rule is useful only if it wins.

11 Limits and Open Problems

The correction of Section 5.3 is the substantive change from an earlier draft, and it is worth stating what it cost. Choice-entropy maximisation is simpler, needs no posterior, and admits a clean combinatorial shortcut. It is also wrong in a way that matters: it cannot tell ignorance from indifference, and a system optimising it drifts toward slates that are maximally ambiguous rather than maximally instructive. Maintaining a posterior over θ is the price of telling those apart, and the simulation of Section 5.3 suggests the two criteria disagree in the large majority of turns.

Several limitations are structural rather than provisional. The candidate pool is generated by the same system whose utilities it trains, so recovered preferences are conditional on one generator's imagination; the divergence constraint bounds this but cannot remove it, and the reserved slot's win rate is the running estimate of what the generator is missing. The recursion (7) requires connectivity that canonicalisation must supply, and the merge threshold supplying it trades directly against the distinctions the instrument exists to measure. The stability of the coupled system of Section 7 is unproven, and the CR-entropy diagnostic proposed there is a

monitor rather than a guarantee. Mutual information is estimated from a finite posterior sample and inherits whatever miscalibration the posterior carries; a confidently wrong posterior yields confidently wrong slates, so calibration diagnostics belong in the deployment monitoring rather than in an appendix.

A privacy asymmetry deserves record. What a person consistently declines is more identifying than what they select: a stable pattern of avoidance across a category supports inference about sensitive attributes without that person entering any sensitive text. Aggregate structure may be publishable where an individual's decline vector is not, and the property making rejection informative to the system makes it disclosive about the person.

12 Conclusion

An answer engine is judged by what it returns. A question engine is judged by what it learns from what it offers.

We have argued that slate selection is an experimental design problem; that the natural objective — maximising the entropy of the predicted choice — confuses what the system does not yet know with what it cannot know, and rewards ambiguity accordingly; and that the correct objective is the mutual information between the preference parameters and the outcome, which prefers slates whose surviving hypotheses disagree. The observations this produces accumulate into a graph whose edges record what people declined, ranked by (7) and fed back through (10) — an asset that compounds with use, appears in no text corpus, and costs a fraction of a cent per observation because it is a by-product of a service people came for.

What remains is the measurement. The deployed system implements the baseline, the instrument records the comparisons, and Section 9 states what would refute the claim. Until those numbers exist this is a design with an argument, which is the honest description of it.

Reproducibility. The simulations of Sections 5.2, 5.3 and 5.6 and the counterexample of Section 5.5 are generated by a single script with fixed seeds; parameters are stated inline at each use. Figures 1–3 and Tables 1 and 3 are computed from the model and are not observations. Table 2 separates measured quantities from those derived under $k = 3$. The reference system why.com is operated by Ockams, Inc.

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Appendix

Proposition 1. $\mathcal{H}(Y | S, x, D) \leq \log_2 k$, with equality iff $\bar{P}$ is uniform on S .

Proof. $\bar{P}$ is a probability vector on k points. Shannon entropy on a k -point support is bounded above by $\log_2 k$, attained uniquely at the uniform distribution [9]. $\square$

Note that under a point-mass posterior, (3) makes $\bar{P}$ uniform exactly when θ is constant on S ; under a dispersed posterior it can also be uniform by cancellation among disagreeing samples, which is the case Section 5.3 exploits.

Proposition 2 (unconstrained only). *Let the posterior be a point mass and order candidates $\theta_1 \geq \dots \geq \theta_n$. Then some slate of size k maximising choice entropy consists of k consecutive indices.*

Proof sketch. Suppose S is optimal and not contiguous. Then there is $j \notin S$ with $\theta_{\min}(S) < \theta_j < \theta_{\max}(S)$. Let S' replace whichever extreme member of S lies further in utility from θ_j . Since (3) is invariant to a common shift of θ , normalise the logits of S to sum to zero; the substitution strictly contracts the spread of the logit vector, so the resulting probability vector is majorised by the original. Entropy is Schur-concave [10], hence $\mathcal{H}(S') \geq \mathcal{H}(S)$. Iterating terminates at a contiguous slate of no lower entropy. $\square$

We verified this numerically over 2×10^5 random unconstrained instances with $n \in [4, 9]$ and $k = 3$, comparing the exhaustive optimum against the best contiguous window; no counterexample was found.

The result does not extend to the constrained problem. Take $\theta = (5, 4, 3, 2, 1)$ and a redundancy ceiling invalidating every adjacent pair. The feasible slates are those with no two adjacent indices, of which $\{1, 3, 5\}$ is the only one; the contiguous windows are $\{1, 2, 3\}$, $\{2, 3, 4\}$ and $\{3, 4, 5\}$, each containing an adjacent pair and therefore infeasible. A window scan returns nothing while a feasible slate exists. Proposition 2 is accordingly stated for the unconstrained case and is not used by Algorithm 1, which enumerates feasible triples exhaustively.